



K-SAFE3D

Korea AI Industrial Safety Platform

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AI Basic Law

7-Tech Fusion

12 AI Agents

3D Digital Twin

Developer: Li Zhixiu (ÇtÉÂÂ)

Kyungwoon University · Dept. of Physical Therapy

China | Korea

1,247

AI Incidents

8.72M

Data Breaches

©3.2T

Economic Loss

12.7%

SME Compliance

Source: Korea AI Safety Institute / Personal Info Protection Commission / Ministry of Trade, 2025

1. Project Background

Korea is facing a rapidly escalating AI industrial safety crisis. Official 2025 government data reveals:

- 1,247 AI safety incidents recorded nationwide (Korea AI Safety Institute)
- 8.72 million individuals affected by AI-related data breaches
- ₩3.2 trillion in industrial economic losses due to AI safety failures
- Only 12.7% of SMEs achieve AI Basic Law compliance (Ministry of Employment)

Existing global AI safety products (OpenAI, Google, NAVER) lack critical capabilities for the Korean market:

- No support for Korean-language attack/defense scenarios
- Not adapted to Korea's AI Basic Law (2024) requirements
- No coverage of Korea's 9-province regional risk scenarios
- No 3D Digital Twin visualization for Korean industrial complexes

2. Our Solution

K-SAFE3D is a Korea-localized AI industrial safety platform developed through customized R&D for the Daegu-Gyeongbuk industrial complex. It fills the critical gap in Korea's AI industrial safety infrastructure.

7: Tech Fusion Architecture

K-AI Korean AI Safety Standard

4 Core Innovations

Interactive 3D visualization of Korea's 9 administrative regions with accident database access. AI Q&A engine for on-demand incident analysis. Automated report generation. Automatically interfaces with Korea's emergency response system. Real-time monitoring of critical infrastructure. Integrated with national disaster response system. Real-time monitoring of critical infrastructure. Integrated with national disaster response system. Real-time monitoring of critical infrastructure. Integrated with national disaster response system.

3D Visualization

Regional DB

AI Q&A

Auto-Plans

Detect

Decide

Execute

Connect

Auto-Report

s, wearable data tampering,

4. Eight Core Features

F1

Real-Time Safety Control Center

AI second-level risk behavior detection with S/A-grade high-risk pop-up alerts and 4 core KPI dashboard.

F2

Field Inspection Simulation

Plan!'Assign!'Field!'Improve!'Verify full digitalization with 9 administrative region inspection logs.

F3

Multi-Dimensional Big Data Analysis

Cross-region/industry/risk-level analysis, monthly risk prediction, auto-generated compliance optimization reports.

F4

3D Metaverse Cyber Range

4 Korean-style AI attack simulations, attack success rate measurement, automatic vulnerability patching.

F5

National 9-Province Accident DB

NAVER/KAKAO/Samsung real accident database with similar-accident risk prediction engine.

F6

AI Basic Law Dual Compliance

Automatic compliance checks, deadline improvement notifications, legal department alerts, full-cycle compliance management.

F7

7-Tech Architecture Comparison

Detailed tech stack breakdown + global 4 competitors horizontal comparison, K-SAFE3D leads in all dimensions.

F8

Safety Gamification + ROI Calculator

Enterprise safety point incentive system + one-click ROI calculation for commercial deployment profitability.

K-SAFE3D

Global Security Operations Center

30%

40%

ROI Analysis — 50-Employee Company

- Average 3 AI incidents per year
- Average loss per incident: €750M
- Annual expected loss without K-SAFE3D: €2.25B
- K-SAFE3D annual subscription: €12M
- Investment payback period: < 4 months
- 3-year ROI: 5,525%

7. Real-World Case Studies

S

Samsung SDI

Region: Gyeonggi-do

Incident: LLM model backdoor attack

Result: ©320M loss prevented

A

NAVER Corp.

Region: Seoul

Incident: AI data exfiltration attempt

Result: 870K personal records protected

B

Daegu SME

Region: Daegu

Incident: Dual compliance violation

Result: Pre-emptive penalty prevention

8. Target Market

- Daegu-Gyeongbuk Industrial Complexes (primary market)
- Korean Small & Medium AI Enterprises (SMEs)
- Korean Local Government Safety Supervision Departments
- National AI regulatory compliance consulting market

9. Contact & Access

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Platform: K-SAFE3D - Korea AI Industrial Safety Platform
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